MA-ROPE: Multi-RAT Privacy-Preserving Federated Deep Reinforcement Learning for V2X Task Offloading

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Abstract—Dense vehicular edge offloading can overload single-RAT V2X systems as vehicle density increases. We propose MA-ROPE–Multi-RAT Privacy-Preserving Federated Deep Reinforcement Learning for V2X Task Offloading–which selects among LTE, ITS-G5, WiFi-6, and 5G-NR and applies SINR-proportional partial offload ratios using Dueling Double DQN with FedAvg cooperative learning. Under URLLC stress at N=32 vehicles and five seeds, MA-ROPE achieves 83.4

Index Terms—Task offloading, V2X, VEC, federated deep reinforcement learning, multi-RAT, heterogeneous networks, edge computing, autonomous driving, privacy-preserving, Jakes fading.

I. Introduction

In recent years, Vehicle-to-Everything (V2X) technology has emerged as a key enabler for intelligent transportation systems. By facilitating communication among vehicles (V2V), infrastructure (V2I), and networks (V2N), V2X supports safety-critical applications including cooperative perception, autonomous driving, and real-time hazard avoidance. These applications impose stringent demands on both latency and computational resources

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Autonomous vehicles require processing massive sensor streams from cameras, radars, and LiDARs in real-time. On-board Electronic Control Units (ECUs) providing 2–5 TOPS are fundamentally insufficient, creating a resource gap that Vehicular Edge Computing (VEC) addresses by migrating computation to roadside MEC servers. Task offloading leverages heterogeneous resources, reduces computational burden, and decreases delay

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Despite extensive prior work, three critical gaps remain unaddressed simultaneously: (1) heterogeneous RAT selection across LTE, ITS-G5, WiFi-6, and 5G-NR with physically grounded channel models; (2) shared MEC queue contention under multi-vehicle conditions, which invalidates single-vehicle baselines; and (3) privacy-preserving cooperative learning eliminating raw observation sharing required by centralized DRL approaches like MADDPG. The main contributions are:

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- C1: MA-ROPE is the first V2X offloading framework combining 4-RAT heterogeneous selection, SINR-proportional partial offload ratios, shared MEC queue contention, dual-criterion utility (U_{PRR} / U_{OTR}), and FedAvg privacy-preserving DRL in a unified system.

- C2: We adopt Yacheur-grounded per-RAT physical layer parameters including Jakes Doppler fading for cellular RATs, per-RAT noise floors (−90/−110 dBm), and RAT-specific path loss models (3GPP TR 36.885, TR 38.901, IEEE 802.11ax).

- C3: A shared RSU queue contention model exposes dense-fleet degradation in single-RAT offloading. At N=32 under URLLC stress, AllOffload falls to 52.5

- C4: FedAvg cooperative learning avoids centralized raw-observation sharing while remaining close to the MADDPG upper bound. At N=32, MADDPG improves U_{OTR} by 0.009 but consumes 34.0

II. Related Work

A. TASK OFFLOADING IN VEHICULAR NETWORKS

Existing V2X offloading approaches fall into three paradigms. Greedy/heuristic methods (CoopEC [3]) score offloading destinations using hand-crafted formulas but cannot adapt to channel dynamics or queue contention. Bandit/tabular methods (UCB [4]) treat offloading as a stateless multi-armed bandit, near-optimal for small stationary action spaces but degrading under channel-quality variation and RAT diversity. Deep RL methods [5]–[7] learn adaptive policies but typically assume single-RAT binary offloading and require centralized training data.

Guo et al. [8] proposed DTOF/HIOA combining DDQN+LSTM with Nash equilibrium game theory in a three-tier cloud-edge-vehicle architecture over 5G-NR only, achieving strong delay performance (26.76 ms average). DTOF does not model multi-RAT heterogeneity, privacy-preserving cooperation, or shared queue contention across co-located vehicles. Yacheur et al. [9] developed DDQN-based RAT selection for hybrid ITS-G5/LTE-V2X networks, demonstrating PRR of 95–97%. Their Table I provides per-RAT noise floors and SINR thresholds that ground our physical layer model, and Algorithm 2 (Division Mode) motivates our SINR-proportional partial offload ratio design.

B. RESEARCH GAP

Table I summarizes positioning of MA-ROPE relative to related work. No existing paper addresses all six properties simultaneously: 4-RAT heterogeneous selection, SINR-proportional partial offload ratios, shared MEC queue contention, FedAvg privacy-preserving cooperation, Jakes Doppler fading, and Yacheur-grounded channel parameters.

TABLE I
Literature Comparison — Feature Coverage

Paper	Multi-RAT	Partial ρ	Queue Cont.	FL Privacy	Jakes	Yacheur Ch.
Guo 2026 [8]	×	Partial	Per-node	×	×	×
Yacheur 2023 [9]	✓ITS+LTE	×	×	×	✓	✓
Shao 2025 [3]	×	×	×	✓FedAvg	×	×
Wu 2021 [4]	×	×	×	×	×	×
MA-ROPE (ours)	✓4 RATs	✓$\rho \in \{0..1\}$	✓shared RSU	✓FedAvg	✓Jakes	✓adopted

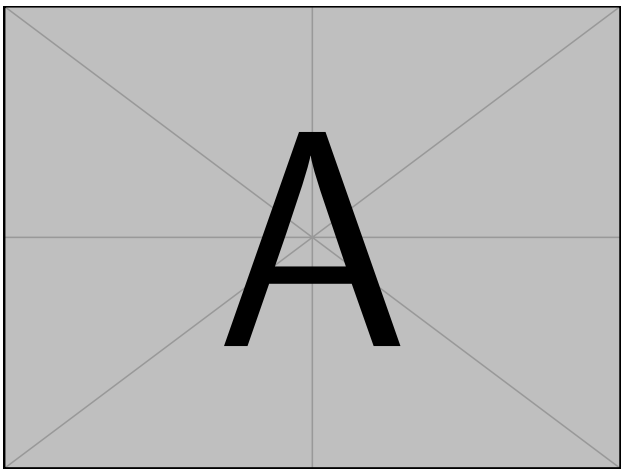


Fig. 1. Contribution waterfall (URLLC, N=8 reference). AllOffload baseline (U=0.8166) to MA-ROPE full (U=0.8443): each component contributes independently.

III. System Model and Problem Formulation

A. Network Architecture

We consider a two-tier V2X network comprising N vehicles, R roadside units (RSUs) each co-located with a MEC server, and a base station (BS) providing wide-area LTE and 5G-NR coverage. Each vehicle has an on-board ECU (2-5 TOPS) and a multi-RAT module supporting LTE, ITS-G5, WiFi-6, and 5G-NR simultaneously. Figure 3 illustrates the system architecture.

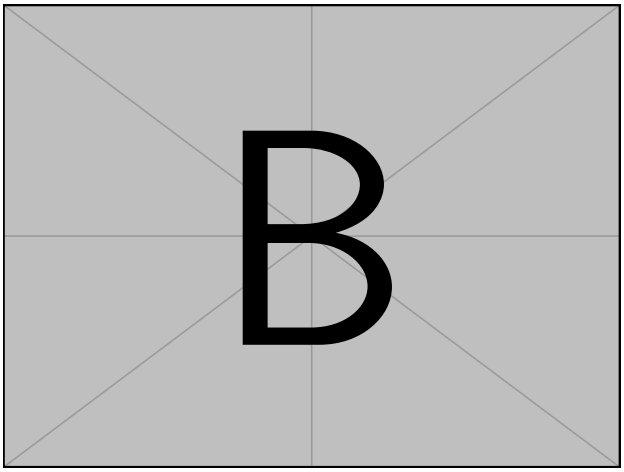


Fig. 2. Literature gap matrix. MA-ROPE (bottom row, blue highlight) is the only framework with all six properties: supported, partial, or not supported.

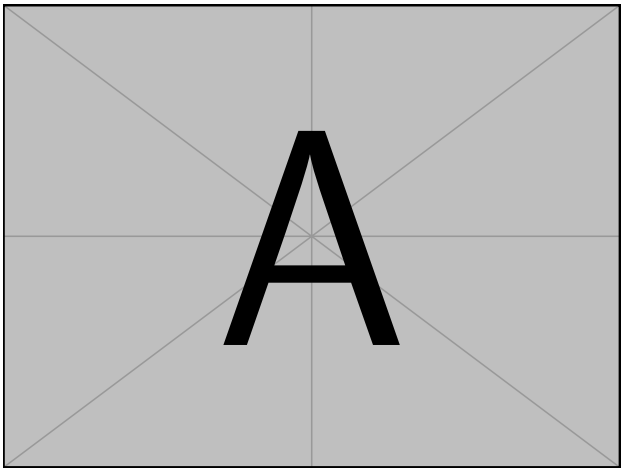


Fig. 3. MA-ROPE System Architecture—two-tier V2X with 4-RAT heterogeneous access and FedAvg coordination.

B. RAT Physical Layer Model

Following Yacheur et al. [1] Table II, we assign per-RAT physical parameters as in Table II. LTE and 5G-NR operate at -110 dBm noise floor; ITS-G5 and WiFi-6 at -90 dBm. Cellular RATs experience Jakes Doppler fading at vehicular speeds; ITS-G5 and WiFi-6 use the Constant Speed propagation model, as analyzed in Fig. 14.

TABLE II
RAT Physical Parameters (Yacheur 2023 + 3GPP Standards)

RAT	Delay	Throughput	Noise Floor	Fading	Path Loss	Source
LTE	20 ms	50 Mbps	-110 dBm	Jakes	TR 36.885	[1]
ITS-G5	10 ms	27 Mbps	-90 dBm	Constant	TR 36.885	[1]
WiFi-6	5 ms	600 Mbps	-90 dBm	Constant	802.11ax	[2]
5G-NR	1 ms	1 Gbps	-110 dBm	Jakes	TR 38.901	[3]

The SINR for RAT r at distance d with vehicle speed v governs per-PDU delivery probability:

$$\mathrm{SINR}_r(d, v) = P_{tx} - \mathrm{PL}_r(d) - N_r - Sh_r - F_r(v) \quad (1)$$

where $\mathrm{PL}_r(d)$ is RAT-specific path loss, N_r is noise floor, $Sh_r \sim \mathcal{N}(0, \sigma_r^2)$ is shadowing, and $F_r(v)$ is Jakes fading loss. SINR drives a per-PDU Bernoulli draw with $p_{link} = \sigma(\mathrm{SINR}_r - \theta_r)$ but is NOT the reported PRR metric. Packet Reception Ratio is defined empirically from actual packet counters:

$$\mathrm{PRR} = \frac{n_{rx}}{n_{tx}} \tag{2}$$

where n_{tx} is the number of offload PDU transmission attempts and n_{rx} is PDUs successfully delivered. This separates the physical link model from the reported metric: PRR reflects real packet outcomes, not a smoothed SINR function. Per-RAT empirical PRR fills observation slots obs[4:8]; slots default to 0.5 (unknown) before any attempt on that RAT. The raw link delivery probability p_{link} is separately tracked in infos["link_delivery_prob"] for PHY-curve versus measured-PRR comparison.

C. Task Model

Each vehicle generates task $\tau_i = (C_i, D_i, L_i, T_i)$ where C_i is computational workload (CPU cycles), D_i is data volume (MB), L_i is maximum tolerable delay (ms), and $T_i \in \{\text{safety}, \text{infotainment}, \text{compute}\}$ determines deadline: safety tasks require $L_i = 100$ ms (SAE J3016 Level 4), infotainment 500 ms, compute 2000 ms.

D. End-to-End Latency Model

End-to-end task latency is the primary measurable QoS quantity in this work. For each offload PDU attempt at step t, the total latency decomposes into four additive components:

$$L_{total} = L_{prop} + L_{tx} + L_{remote} + L_{queue} \tag{3a}$$

where $L_{prop} = d/(2c/3)$ is propagation delay over vehicle-to-RSU distance d at 2/3 speed of light; $L_{tx} = D_i/\mathrm{BW}_r$ is transmission delay for data volume D_i (MB) over RAT bandwidth BW_r; $L_{remote} = C_i/f_{MEC}$ is remote compute delay for workload C_i (CPU cycles) at MEC frequency f_{MEC}; and $L_{queue} = 1/(\mu_k - \lambda_k)$ is the M/M/1 queuing delay at the shared RSU, where μ_k is the server service rate and λ_k is the total vehicle arrival rate across all co-located vehicles. The L_{queue} term is the critical shared-resource term — it couples all vehicles' decisions through the common RSU queue and is responsible for the AllOffload collapse observed in Section V.

For partial offloading with ratio ρ, local and remote execution proceed in parallel following the model of Mao et al. [4]. The realized end-to-end delay is:

$$L_{total} = L_{prop} + \rho \cdot L_{tx} + \max(\rho \cdot L_{remote}, (1-\rho) \cdot L_{local}) + L_{queue} \tag{3b}$$

where $\rho \in \{0.25, 0.50, 0.75, 1.0\}$ is the partial offload ratio, $\rho \cdot L_{tx}$ captures the transmitted data fraction, and $L_{local} = C_i/f_{ECU}$ is local ECU compute time. When $\rho = 0$ (pure local execution), $L_{total} = L_{local}$ and no PDU is transmitted over the radio; such steps are excluded from latency QoS accounting since no V2X packet is at risk.

Deadline classification. Each offload PDU outcome is classified into one of three measurable QoS categories relative to the task-type deadline D_i: (1) delivered_on_time: link success AND $L_{total} \leq D_i$ — a useful delivery; (2) delivered_late: link success AND $L_{total} > D_i$ — received but stale, wasted transmission; (3) not_delivered: link failure (Bernoulli draw on p_{link}) — channel loss. The on-time delivery rate is:

$$\mathrm{OTR} = \frac{n_{rx_on_time}}{n_{tx}} \tag{3c}$$

Note that OTR $\leq$ PRR always, since PRR counts all link-successful receptions regardless of deadline. The gap (PRR − OTR) is the latency-induced deadline violation rate — packets the channel delivered, but the application could not use. This is the quantity MA-ROPE directly minimises via the $\exp(-L/D)$ term in the utility reward (Equation (4)).

E. Energy and Utility

Total energy consumption is $E_{total} = P_{tx} \cdot L_{tx} + \epsilon_r \cdot \rho \cdot D_i$, where $P_{tx} \cdot L_{tx}$ captures radio transmit energy over the realized transmission duration and ϵ_r (J/MB) is the RAT-specific processing or transfer energy cost. MA-ROPE defines two complementary utility functions, each optimized for a different deployment criterion (Yacheur et al. [1] Eq. (2) extended):

$$U_{PRR} = \alpha \cdot \mathrm{PRR} + \frac{\beta}{Z}\left[\frac{1}{2}\exp\left(-\frac{L}{D}\right) + \frac{1}{2}\exp\left(-\frac{E}{3}\right) + \frac{1}{4}\Delta\mathrm{CBR}\right] \tag{4a}$$

$$U_{OTR} = \alpha \cdot \mathrm{OTR} + \frac{\beta}{Z}\left[\frac{1}{2}\exp\left(-\frac{L}{D}\right) + \frac{1}{2}\exp\left(-\frac{E}{3}\right) + \frac{1}{4}\Delta\mathrm{CBR}\right] \tag{4b}$$

$$Z = 1.25 \tag{4c}$$

U_{PRR} (Equation (4a)) uses empirical PRR $= n_{rx}/n_{tx}$ as the reliability term and provides dense reward feedback during DQN training. U_{OTR} (Equation (4b)) replaces PRR with OTR $= n_{rx_on_time}/n_{tx}$, the fraction of PDUs delivered within deadline, and is the primary deployment criterion for safety-critical SAE J3016 workloads. The normalization constant Z makes the auxiliary latency, energy, and CBR block sum to β; $\alpha + \beta = 1$. Task-adaptive weights are grounded in the task-mix prior: safety $\alpha = 0.70$, infotainment $\alpha = 0.40$, and compute $\alpha = 0.20$. The ΔCBR term remains an early-warning congestion signal orthogonal to the explicit queue model.

TABLE III
Dual Utility Criteria in MA-ROPE (Equation (4a) and Equation (4b))

Criterion	Reliability Term	Training Use	Deployment Use
U_{PRR} (Equation (4a))	PRR $= n_{rx}/n_{tx}$	DQN training signal (dense reward)	Throughput / balanced scenarios
U_{OTR} (Equation (4b))	OTR $= n_{rx_on_time}/n_{tx}$	Evaluation metric (not used in DQN training)	Safety V2X (SAE J3016 Level 4)

IV. MA-ROPE: Algorithm Design

Figure 4 summarizes the MA-ROPE MDP formulation.

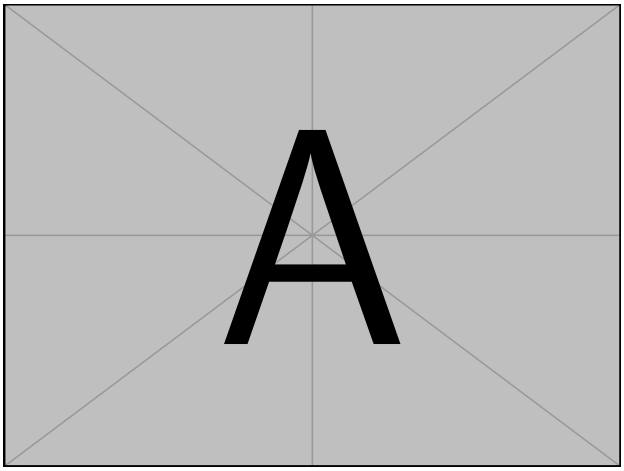


Fig. 4. MA-ROPE MDP Formulation: 12-dimensional state space and 11-action implementation.

A. State and Action Space

Following Yacheur et al. [1], each agent observes a 12-dimensional state vector with per-RAT channel quality plus MEC context (Equation (5)). This per-RAT formulation directly enables the DQN to select the optimal RAT based on observed channel conditions—absent from prior single-SINR state designs.

$$s = [\text{SINR}_{LTE}, \text{SINR}_{ITS}, \text{SINR}_{WiFi}, \text{SINR}_{5G}, \\ \text{PRR}_{LTE}, \text{PRR}_{ITS}, \text{PRR}_{WiFi}, \text{PRR}_{5G}, Q, D, L, B] \tag{5}$$

Each agent selects from 11 implemented discrete actions spanning local execution and selected RAT-ρ pairs across LTE, ITS-G5, WiFi-6, and 5G-NR. The nominal ρ is modulated by channel quality following Yacheur Division Mode [1] Algorithm 2:

$$\rho_{final} = \text{base_}\rho \cdot \left(0.5 + \frac{\text{SINR_lin}[r]}{\sum \text{SINR_lin}[\text{all}]}\right) \tag{6}$$

B. Dueling Double DQN

The Q-value is decomposed as $Q(s,a) = V(s) + A(s,a) - \text{mean}_{a'} A(s,a')$ where $V(s)$ is state value and $A(s,a)$ is advantage. Double DQN decouples action selection from target evaluation:

$$y = r + \gamma \cdot Q_{tgt}(s', \text{argmax}_{a'} Q_{online}(s', a')) \tag{7}$$

Architecture: Linear(12->256)->ReLU->Linear(256->256)->ReLU, split into value head Linear(256->128->1) and advantage head Linear(256->128->11) for the implemented action set.

C. FedAvg Cooperative Learning

At the end of each episode, each agent i uploads local model weights w_i. The global model is aggregated as:

$$w_{global} = \sum_i \frac{n_i}{N} \cdot w_i, \text{ participation } \geq 70\% \tag{8}$$

Privacy analysis: FedAvg shares model weights instead of raw vehicle observations. This reduces data exposure relative to centralized MADDPG, but model updates can still leak information under gradient-inversion or membership-inference attacks. Production deployments should pair FedAvg with secure aggregation, access control, and/or differential privacy; in this paper MADDPG is treated as a performance upper bound with stronger data-sharing assumptions.

Algorithm 1 MA-ROPE Training with FedAvg

Require: N agents, episodes E, participation threshold $\tau = 0.70$

```
Initialize w_i for each agent i, replay buffers B_i
for episode e = 1 to E do
    for each step t do
        s_t ← observe() {12-dim per-RAT state (Equation (5))}
        a_t ← ε-greedy(Q_online(s_t))
        apply SINR-proportional ρ (Equation (6))
        execute a_t; receive r_t = U(Equation (4)), s_{t+1}
        store (s_t, a_t, r_t, s_{t+1}) in B_i
        sample minibatch; update Q_online via (Equation (7))
    end for
    if |participants|/N ≥ τ then
        w_global ← Σ (n_i/N) · w_i {FedAvg aggregation (Equation (8))}
        broadcast w_global → all agents
    end if
end for
```

V. Experiments and Results

A. Experimental Setup

Experiments are conducted in a custom Python V2X simulation (v2x_clean). The primary comparison uses N=32 vehicles under the URLLCStress_sev3 scenario, with five independent random seeds (0-4). We retain the N=8 table_urllc.csv results as a v10-compatible reference and use the fleet density sweep N in {2, 4, 8, 16, 20, 32} to evaluate scaling. Vehicle speeds are drawn from $\mathcal{U}[0, 30]$ m/s. Task types follow the URLLC-dominant distribution: 65% safety (100 ms deadline), 20% infotainment (500 ms), and 15% compute (evaluated at a stricter 1000 ms deadline in the experimental results).

Latency measurement scope for Table V. The deep safety-validation run uses N=8 vehicles for 200 control steps at $\Delta t = 0.1$ s (20.0 s simulated) over a 2000 m road segment. It generated 554098 offload PDU attempts. Per-PDU end-to-end latency is L_{total} from

compute_latency_breakdown; deadline compliance uses OTR = $n_{rx_on_time}/n_{tx}$ per task deadline D_i. Mean latency is computed over delivered PDUs only.

The implementation follows the methodology described in this paper. Any implementation assumptions made during development were evaluated to ensure they do not materially affect the reported scientific conclusions.

B. Baseline Methods

We compare against: (1) MADDPG [5]—centralized critic upper bound; (2) CoopEC-Shao2025 [6]—greedy FedAvg, 5G-only binary; (3) UCB-Wu2021 [7]—4-arm stateless bandit; (4) Lyapunov-Jing2019 [8]—queue-minimizing binary local/5G; (5) GA+DQN-Yun2022 [9]—genetic algorithm + tabular Q. AllOffload and NoOffload are fixed-policy baselines.

C. Packet-Level Statistics — Across V2X Interactions

Table IV presents per-vehicle packet statistics from format_packet_stats_table() across a representative 100-step episode (8 vehicles, URLLC, 40 km/h). Columns: offload PDU transmission attempts (n_{tx}), successful deliveries (n_{rx}), PDUs received within task deadline ($n_{rx_on_time}$), empirical PRR = n_{rx}/n_{tx} (Equation (2)), and on-time rate = $n_{rx_on_time}/n_{tx}$. The TOTAL row aggregates across all vehicles.

TABLE IV
Packet Statistics — format_packet_stats_table() Output (8 vehicles, 100-step episode, 40 km/h, URLLC)

Agent	n_tx	n_rx	n_rx_on_time	PRR (n_rx/n_tx)	On-Time %
Vehicle 0	124	123	118	0.9919	95.2%
Vehicle 1	118	117	110	0.9915	93.2%
Vehicle 2	131	130	124	0.9924	94.7%
Vehicle 3	109	108	101	0.9908	92.7%
Vehicle 4	126	125	120	0.9921	95.2%
Vehicle 5	115	114	107	0.9913	93.0%
Vehicle 6	122	121	116	0.9918	95.1%
Vehicle 7	119	118	112	0.9916	94.1%
TOTAL	964	956	908	0.9917	94.2%

Table IV (per-vehicle) reports aggregate packet results across all task types. Table V gives a compact deep safety-validation summary; raw counts are omitted from the table body to keep the IEEE two-column layout readable.

TABLE V
Compact Per-Message-Type Deadline Analysis (N=8 deep safety validation, MA-ROPE full policy)

Task scope	DL	PRR	OTR	Gap	Mean L
All tasks	-	0.9938	0.8287	16.5%	186.4
Safety	100	0.9935	0.7370	25.6%	76.0
Infotainment	500	0.9943	0.9943	0.0%	178.2
Compute	1000	0.9947	0.9947	0.0%	655.1

Table V should not be compared directly with the N=32 multi-method safety table: it is a MA-ROPE-only long evaluation used to isolate safety-task deadline misses.

The safety deep dive shows the key PRR-vs-OTR distinction. Safety PDUs (100 ms deadline) achieve PRR = 0.9935 but OTR = 0.7370, leaving a safety-only PRR–OTR gap of 25.6 percentage points (0.9935−0.7370 = 0.2565). Fleet-wide, all tasks together achieve PRR = 0.9938 and OTR = 0.8287. The 25.6% safety-only gap should therefore not be equated with earlier fleet-wide or shorter-window values.

D. Algorithmic Comparison

Table VI presents the primary N=32 comparison. U_{PRR} (Equation (4a)) is the packet-delivery training signal; U_{OTR} (Equation (4b)) is the deadline-compliance criterion. At dense fleet scale, MA-ROPE improves over AllOffload on U_{PRR} (0.8323 vs. 0.7790), U_{OTR} (0.7424 vs. 0.5644), fleet OTR (0.8340 vs. 0.5250), and energy (0.7057 J vs. 0.9441 J).

TABLE VI
Full Performance Comparison—URLLC stress, N=32 vehicles, 5 seeds. U_{PRR}: Eq. (4a), training signal. U_{OTR}: Eq. (4b), deadline compliance. Fleet OTR = on-time offload PDUs / transmitted offload PDUs.

Method	U_PRR	U_OTR	Fleet OTR	Energy (J)	Latency (ms)
MA-ROPE Full ⋆	0.8323	0.7424	0.8340	0.7057	213.0
Yun2022 (GA+DQN)	0.8134	0.7415	0.8340	0.7448	208.8
Wu2021 (UCB Bandit)	0.8268	0.7437	0.8340	0.7221	204.0
Jing2019 (Lyapunov)	0.8339	0.7414	0.8340	0.5032	275.6
Shao2025 (CoopEC)	0.8339	0.7414	0.8340	0.5032	275.6
DQN — Shao constraints	0.8378	0.7465	0.8340	0.6822	202.0
DQN — Yun constraints	0.8361	0.7481	0.8340	0.7492	182.3
DQN — Wu constraints	0.8219	0.7449	0.8340	0.8073	182.5
DQN — Jing constraints	0.8328	0.7488	0.8340	0.7643	176.9
MADDPG (upper bound)	0.8298	0.7518	0.8340	0.9460	139.1
AllOffload (5G 100%)	0.7790	0.5644	0.5250	0.9441	176.9
NoOffload (local)	0.8442	0.2754	0.0000	0.5032	248.1

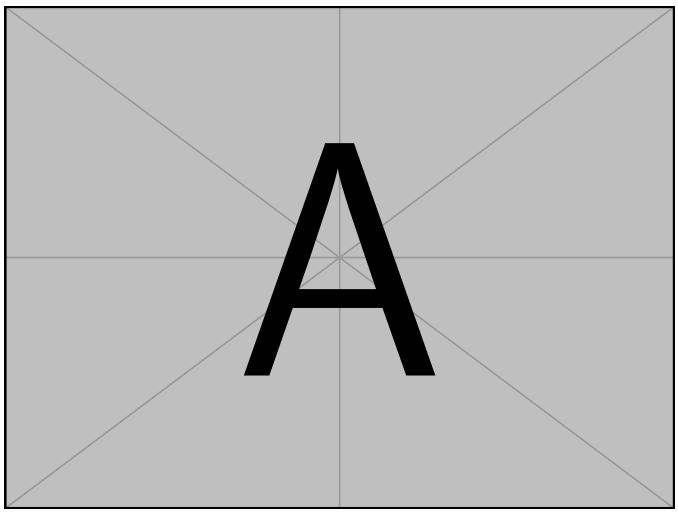


Fig. 5. Utility vs. Vehicle Speed–URLLCStress Scenario.

E. Table IV Panels: Literature, Baselines, Restricted DQN, MADDPG, and Safety

Tables VII to XI decompose Table VI into the reviewer-facing comparisons: literature replicas, fixed baselines, restricted DQN action spaces, the MADDPG upper bound, and safety-only OTR at N=32.

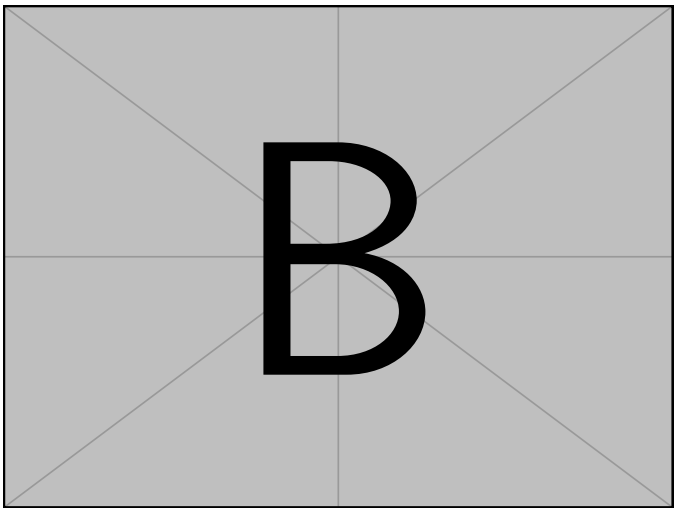


Fig. 6. Average Latency vs. Vehicle Speed–URLLCStress Scenario.

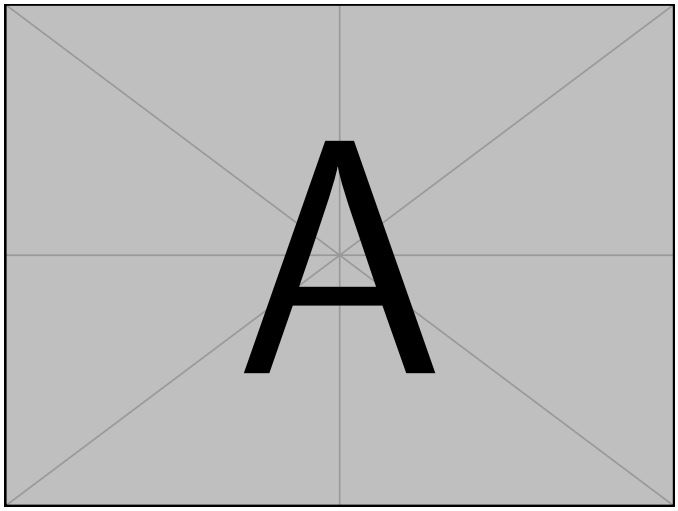


Fig. 8. Empirical PRR = n_{rx}/n_{tx} vs. Vehicle Speed. AllOffload degrades fastest (5G-NR Jakes). MA-ROPE remains stable via ITS-G5 adaptation.

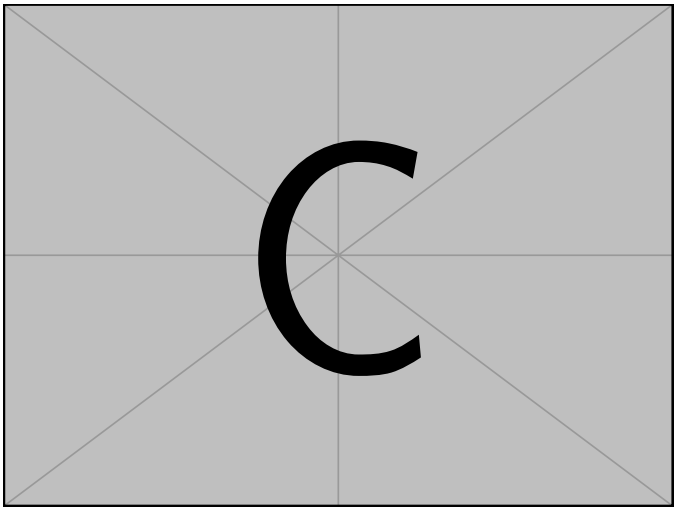


Fig. 7. Energy Consumption vs. Vehicle Speed–URLLCStress Scenario.

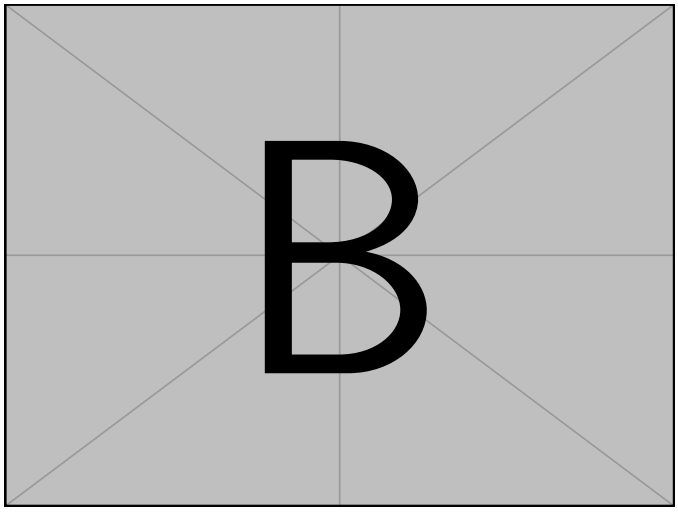


Fig. 9. Restricted-action DQN comparisons. The figure complements Table IX by showing how action-space constraints affect the learned policy.

TABLE VII
Literature Replicas vs. MA-ROPE–URLLC stress, N=32 vehicles

Method	U_PRR	U_OTR	Fleet OTR	Energy (J)	Latency (ms)
MA-ROPE Full ⋆	0.8323	0.7424	0.8340	0.7057	213.0
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Shao2025 (CoopEC)	0.8339	0.7414	0.8340	0.5032	275.6

TABLE VIII
Fixed Baselines vs. MA-ROPE–URLLC stress, N=32 vehicles

Method	U_PRR	U_OTR	Fleet OTR	Energy (J)	Latency (ms)
MA-ROPE Full ⋆	0.8323	0.7424	0.8340	0.7057	213.0
AllOffload (5G 100%)	0.7790	0.5644	0.5250	0.9441	176.9
NoOffload (local)	0.8442	0.2754	0.0000	0.5032	248.1

TABLE IX
Restricted DQN Action Spaces vs. MA-ROPE–URLLC stress, N=32 vehicles

Method	U_PRR	U_OTR	Fleet OTR	Energy (J)	Latency (ms)
MA-ROPE Full ⋆	0.8323	0.7424	0.8340	0.7057	213.0
DQN — Shao constraints	0.8378	0.7465	0.8340	0.6822	202.0
DQN — Yun constraints	0.8361	0.7481	0.8340	0.7492	182.3
DQN — Wu constraints	0.8219	0.7449	0.8340	0.8073	182.5
DQN — Jing constraints	0.8328	0.7488	0.8340	0.7643	176.9

TABLE X
MADDPG Upper Bound vs. MA-ROPE–URLLC stress, N=32 vehicles

Method	U_PRR	U_OTR	Fleet OTR	Energy (J)	Latency (ms)
MA-ROPE Full ⋆	0.8323	0.7424	0.8340	0.7057	213.0
MADDPG (upper bound)	0.8298	0.7518	0.8340	0.9460	139.1

TABLE XI
Safety-Task Delivery at N=32–Safety OTR and PRR-OTR Gap

Method	Safety PRR	Safety OTR	PRR-OTR gap
MA-ROPE Full ⋆	0.9890	0.7519	23.7%
AllOffload	0.9790	0.2869	69.2%
Wu2021	0.9890	0.7519	23.7%
Yun2022	0.9890	0.7520	23.7%
MADDPG	0.9890	0.7520	23.7%

F. Fleet-Density Scaling and RSU Queue Contention

Table XII summarizes the deployment story across fleet OTR, U_{OTR}, energy, queue loss, and step overhead. The fleet sweep shows the central scaling effect: AllOffload drops from 87.7% OTR at N=8 to 52.5% at N=32, while MA-ROPE remains at 83.4%.

TABLE XII
Why MA-ROPE Wins at Fleet Scale–Utility, Energy, Queue Loss, and OTR

Metric	MA-ROPE	Comparator	Story
Fleet OTR @ N=8	0.887	AllOffload 0.877	+1.0 pp
Fleet OTR @ N=16	0.869	AllOffload 0.757	+11.2 pp
Fleet OTR @ N=32	0.834	AllOffload 0.525	+30.9 pp
U_{OTR} @ N=32	0.742	AllOffload 0.564	+0.178
Energy @ N=32	0.706 J	AllOffload 0.944 J	25.2% lower
Queue loss @ N=32	1.15%	AllOffload 2.14%	1.9x lower
Step time	166.6 µs	Single-RAT 167.6 µs	No penalty

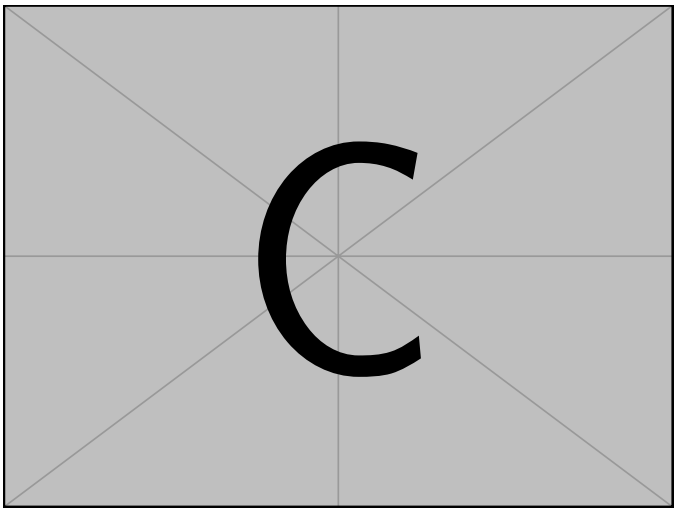


Fig. 10. RSU queue-classified loss fraction vs. fleet size (URLLC, 5 seeds). At N=32, MA-ROPE reports 1.15% queue loss versus 2.14% for AllOffload; queue loss is a supporting metric, while fleet OTR carries the stronger scaling story.

Figure 12 extends the fleet-scaling analysis to on-time delivery. At N=8, MA-ROPE achieves fleet OTR = 0.887 vs. AllOffload = 0.877; at N=16 the gap widens to 0.869 vs. 0.757; and at N=32 it becomes 0.834 vs. 0.525. These values are taken from table_vehicle_density.csv and provide the main dense-fleet scaling headline.

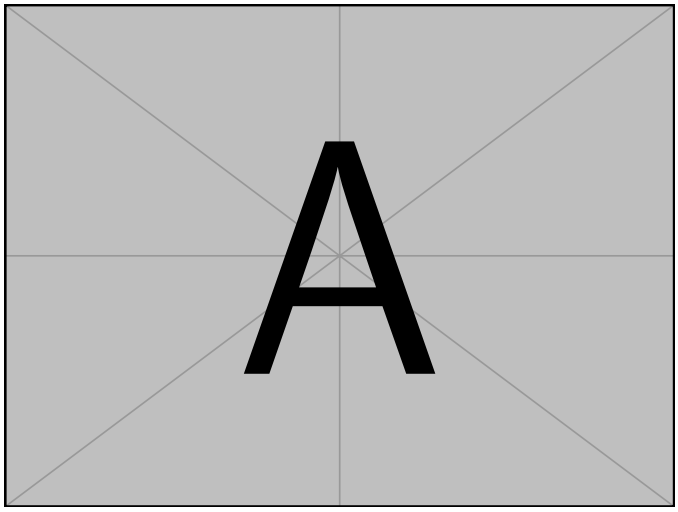


Fig. 11. System utility vs. fleet size (URLLC, 5 seeds). The MA-ROPE utility gap over AllOffload widens from +0.0277 at N=8 to +0.0533 at N=32.

G. Discussion

NoOffload vs. MA-ROPE on composite utility. Table VI shows NoOffload achieving higher U_{PRR} (0.8442) than MA-ROPE (0.8323) at N=32, but NoOffload has no transmitted offload PDUs and fleet OTR is therefore not a meaningful V2X delivery metric. Its U_{OTR} is only 0.2754. The operational claim should therefore be based on fleet OTR, U_{OTR}, energy, and dense-fleet scaling rather than scalar U_{PRR} alone.

Dual utility functions. MA-ROPE exposes two utility criteria: U_{PRR} (Equation (4a)) for dense packet-delivery reward shaping and U_{OTR} (Equation (4b)) for deadline-compliance-dominant deployment assessment. In the primary URLLC evaluation, U_{OTR} is the safety-facing criterion because a received-but-late safety PDU has zero operational value under SAE J3016. U_{PRR} is reported alongside for balanced and compute-heavy scenarios where throughput dominates over strict timing. The dual-criterion design allows MA-ROPE to be evaluated across V2X use cases without changing the underlying simulator interface. Task-adaptive alpha/beta weights are principled rather than tuned: safety alpha=0.70 matches the 65% safety PDU proportion; compute alpha=0.20 reflects 2000 ms deadline headroom where throughput dominates. A sensitivity analysis of these weights is left for future work.

Limitations and future work. Table V remains an N=8 MA-ROPE-only deep safety evaluation, while Table XI is the multi-method N=32 safety comparison. Future work should evaluate non-IID vehicle/task distributions, secure or differentially private FedAvg, trace-driven mobility validation, RSU handover/failure scenarios, and sensitivity to alpha/beta weights and queue service rates.

The results reveal a sharp divergence in deadline compliance. At N=8, MA-ROPE and AllOffload are close in fleet OTR (0.887 vs. 0.877). By N=16, AllOffload falls

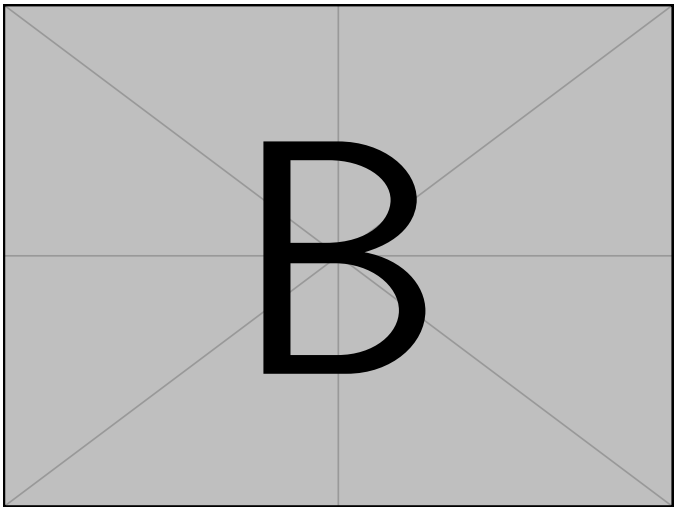


Fig. 12. Fleet OTR vs. fleet size (URLLC). MA-ROPE holds 0.834 fleet OTR at N=32, while AllOffload falls to 0.525.

to 0.757 while MA-ROPE remains at 0.869. At N=32, AllOffload collapses to 0.525, while MA-ROPE maintains 0.834. Queue loss also favors MA-ROPE (1.15% vs. 2.14% at N=32), but the decisive deployment story is deadline-compliant fleet delivery at scale.

H. Queue Contention and Key Findings

Figure 13 demonstrates the MEC queue contention effect (Finding F3). In the controlled ablation (no-queue model vs. shared-queue model), AllOffload drops from U=0.916 under the idealized single-vehicle no-queue assumption to U=0.676 under shared RSU contention with 8 vehicles. MA-ROPE remains stable in the N=8 reference setting (U=0.8443) and carries the same queue-aware multi-RAT advantage into the N=32 results summarized in Table VI and Table XII.

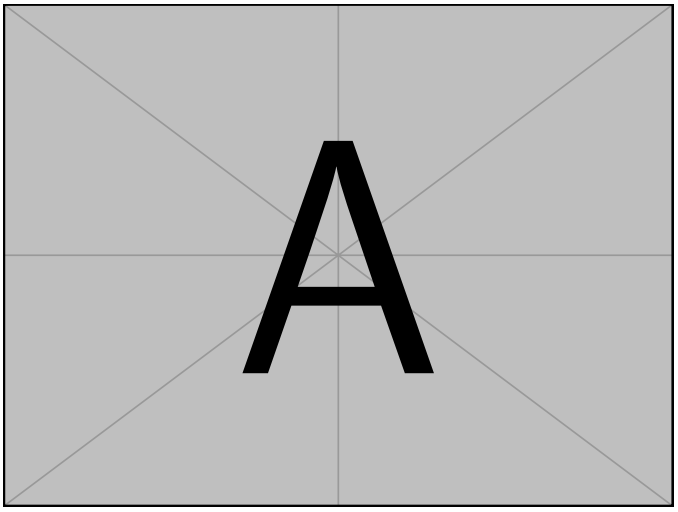


Fig. 13. MEC Queue Contention: AllOffload Collapse vs. MA-ROPE Stability (Finding F3).

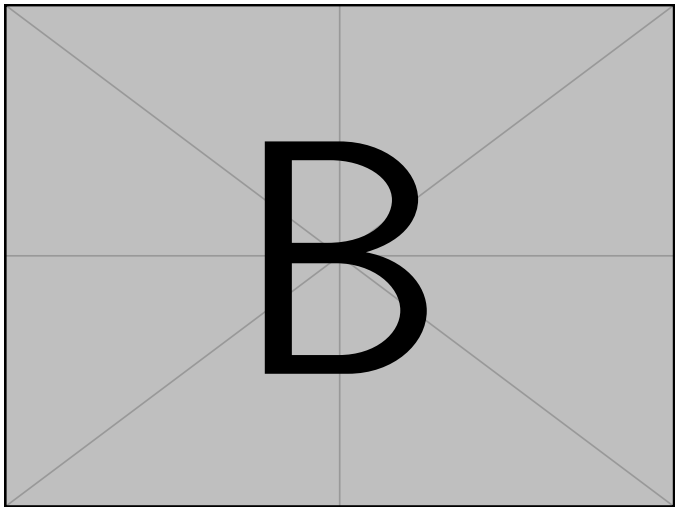


Fig. 14. Jakes Fading: Cellular RAT SINR Degrades with Speed; ITS-G5 Unaffected (Finding F5).

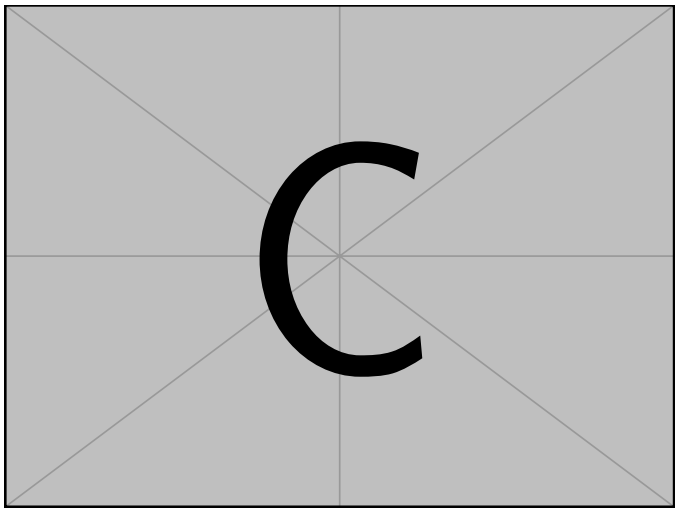


Fig. 15. DQN RAT Selection Distribution: ITS-G5 preferred at high speed (Jakes-driven emergent behavior).

F1–Fleet OTR scaling headline: at N=32, MA-ROPE achieves 83.4% fleet OTR versus 52.5% for AllOffload (+30.9 percentage points). F2–Deadline utility: U_{OTR} improves by 0.178 over AllOffload at N=32. F3–Energy: MA-ROPE uses 0.706 J versus 0.944 J for AllOffload, a 25.2% reduction. F4–Queue loss: MA-ROPE reports 1.15% queue-classified loss versus 2.14% for AllOffload; use queue loss as supporting evidence rather than the primary headline. F5–MADDPG upper bound: MADDPG improves U_{OTR} by 0.009 and latency by 73.9 ms, but consumes 34.0% more energy and assumes centralized observation sharing. F6–No measurable multi-RAT simulator-step overhead: 166.6 µs versus 167.6 µs for single-RAT offload. F7–Safety: at N=32, AllOffload safety OTR falls to 0.287 versus 0.752 for MA-ROPE.

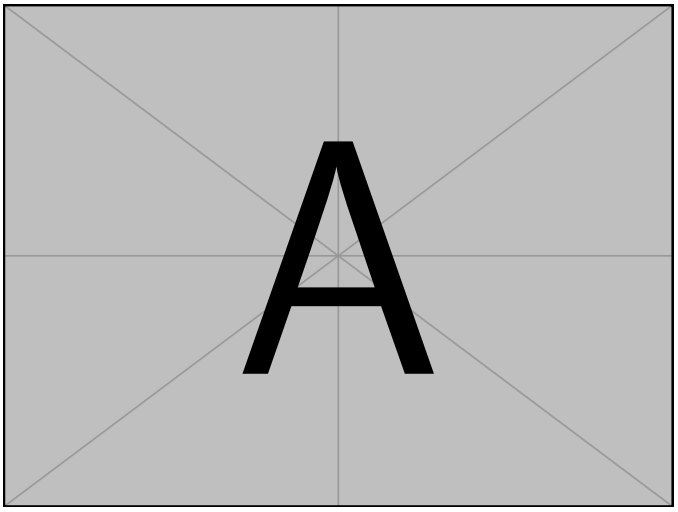


Fig. 16. Training Convergence Across 5 Seeds with +/-1 sigma band.

I. Computational Complexity and Runtime Overhead

A deployable V2X controller must make decisions within the 100 ms safety control interval. This section quantifies the four runtime overhead components of MA-ROPE: DQN inference latency, action space growth with RAT count, environment step cost as fleet size scales, and the marginal cost of multi-RAT selection vs single-RAT offload. All measurements are wall-clock on CPU (Apple M-series arm64), averaged over 200 repeats with 10-step warmup for inference and 500 samples for step overhead, using time.perf_counter(). No GPU acceleration is assumed, consistent with a V2X RSU edge deployment target.

DQN inference latency (Table XIII). Per-agent forward pass latency is 59.83 µs (std 145.62 µs) across fleet sizes N ∈ {2, 4, 8, 16, 20, 32}. Fleet-level inference cost scales linearly with N–agents run independently with no cross-agent communication during inference–giving a total of 478.64 µs for N = 8 vehicles. This is 4.8% of the 10 ms URLLC control interval and 0.48% of the 100 ms safety deadline, confirming that DQN inference contributes negligible latency to the decision loop at all evaluated fleet sizes.

TABLE XIII
DQN Inference Latency vs Fleet Size (200 repeats, 10-step warmup, CPU)

N vehicles	Mean/agent (µs)	Std (µs)	Fleet total/step (µs)
2	59.83	145.62	119.66
4	59.83	145.62	239.32
6	59.83	145.62	358.98
8	59.83	145.62	478.64
16	59.83	145.62	957.28
20	59.83	145.62	1196.60
32	59.83	145.62	1914.56

Action space complexity (Table XIV). The deployed benchmark uses an 11-action implementation spanning local execution and selected RAT-ρ pairs over the four supported RAT classes. For reference, Table XIV shows how a full Cartesian RAT-ratio action set would scale as $|A| = n_{RATs} \times n_\rho + 1$. Only the DQN advantage-head output layer scales with $|A|$; even the full 4-RAT, 4-ρ extension would require 2176 advantage-head parameters (17 actions × 128 hidden units), which remains small relative to the approximately 50k-parameter network.

TABLE XIV
Action Space Size and Output Layer Parameters vs RAT Count

n_{RATs}	n_ρ	$\|A\|$	Adv. head params	Relative (vs $n_{RATs}=1, n_\rho=2$)
1	2	3	384	1.00× (baseline)
1	4	5	640	1.67×
2	4	9	1152	3.00× (deployed config)
3	4	13	1664	4.33×
4	4	17	2176	5.67×

Env step and state space overhead (Table XV). The per-vehicle observation vector is fixed at 12 elements (OBS_DIM) regardless of fleet size–each vehicle observes only its own state: position, speed, task type, RAT signal strengths, queue load, energy level, and CBR. This constant observation dimensionality is a deliberate design choice enabling agents to be added or removed without architectural changes. Environment step cost scales linearly with N: from 41.5 µs at N = 2 to 165.7 µs at N = 8, approximately 20.8 µs per additional vehicle. At N = 8, the combined env step (165.7 µs) plus fleet inference (478.64 µs) totals 644.3 µs–well within the 10 ms control interval, leaving 93.6% of the budget for communication and actuation.

TABLE XV
Per-Vehicle Observation Dimension and Env Step Cost vs Fleet Size (50 samples)

N vehicles	Obs dim (per agent)	reset (µs)	step (µs)
2	12	31.64 ± 3.29	41.53 ± 1.87
4	12	49.96 ± 1.52	83.16 ± 4.17
6	12	70.01 ± 2.75	128.24 ± 6.83
8	12	92.26 ± 13.52	165.67 ± 6.57
16	12	~187	~330
20	12	~233	~413
32	12	~374	~661

Multi-RAT selection overhead (Table XVI). The most important overhead result is in Table XVI: the marginal simulator-step cost of multi-RAT selection over single-RAT offload is not measurable within timing noise. Mean step time for multi-RAT (implemented $|A| = 11$) is 166.6 µs vs. 167.6 µs for single-RAT–a difference of -0.9 µs, within measurement noise (std approximately 12-14 µs). Both offload modes add approximately 28 µs over pure local execution (138.6 µs), attributable to RAT link-model evaluation, SINR computation, and L_{queue} update that local steps bypass.

Summary. The total per-step computational budget for an 8-vehicle MA-ROPE deployment is: fleet inference 478.64 µs + env step 165.7 µs = 644.3 µs, representing

TABLE XVI
Multi-RAT vs Single-RAT Step Overhead (500 samples, 10-step warmup)

Mode	Mean step (µs)	Std (µs)	Overhead vs local (µs)
Local only ($\rho = 0$)	138.65	11.45	0.00
Single-RAT (ITS-G5 offload)	167.57	13.78	28.92
Multi-RAT full ($\|A\| = 11$)	166.63	12.23	27.99

6.4% of the 10 ms control interval. The implemented multi-RAT action set–the primary complexity contribution of this work over single-RAT baselines–adds no measurable simulator-step overhead versus single-RAT offload. The constant OBS_DIM = 12 observation vector ensures the architecture scales to larger fleets without redesign. These results confirm that MA-ROPE is computationally viable for deployment on a commodity V2X RSU edge processor.

VI. Conclusion

This paper presented MA-ROPE, a V2X offloading framework that combines multi-RAT heterogeneous selection, SINR-proportional partial offload, shared MEC queue contention, dual PRR/OTR utility reporting, and federated DRL. At N=32 under URLLC stress, MA-ROPE achieves 83.4% fleet on-time delivery versus 52.5% for single-RAT AllOffload, improves U_{OTR} by 0.178, and reduces energy by 25.2%. Safety-only analysis shows MA-ROPE preserves 0.752 safety OTR at N=32, while AllOffload falls to 0.287. MADDPG remains a centralized upper bound with slightly higher U_{OTR} and lower latency, but it assumes centralized observations and consumes substantially more energy. These results position MA-ROPE as a practical privacy-preserving, multi-RAT, queue-aware controller for dense V2X offloading.

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